

Convolution and Deconvolution



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Convolution means several things:

- IS multiplication of a polynomial series
- IS a mathematical process
- IS filtering

Convolution means several things:

- IS multiplication of a polynomial series

$$A * B = C$$

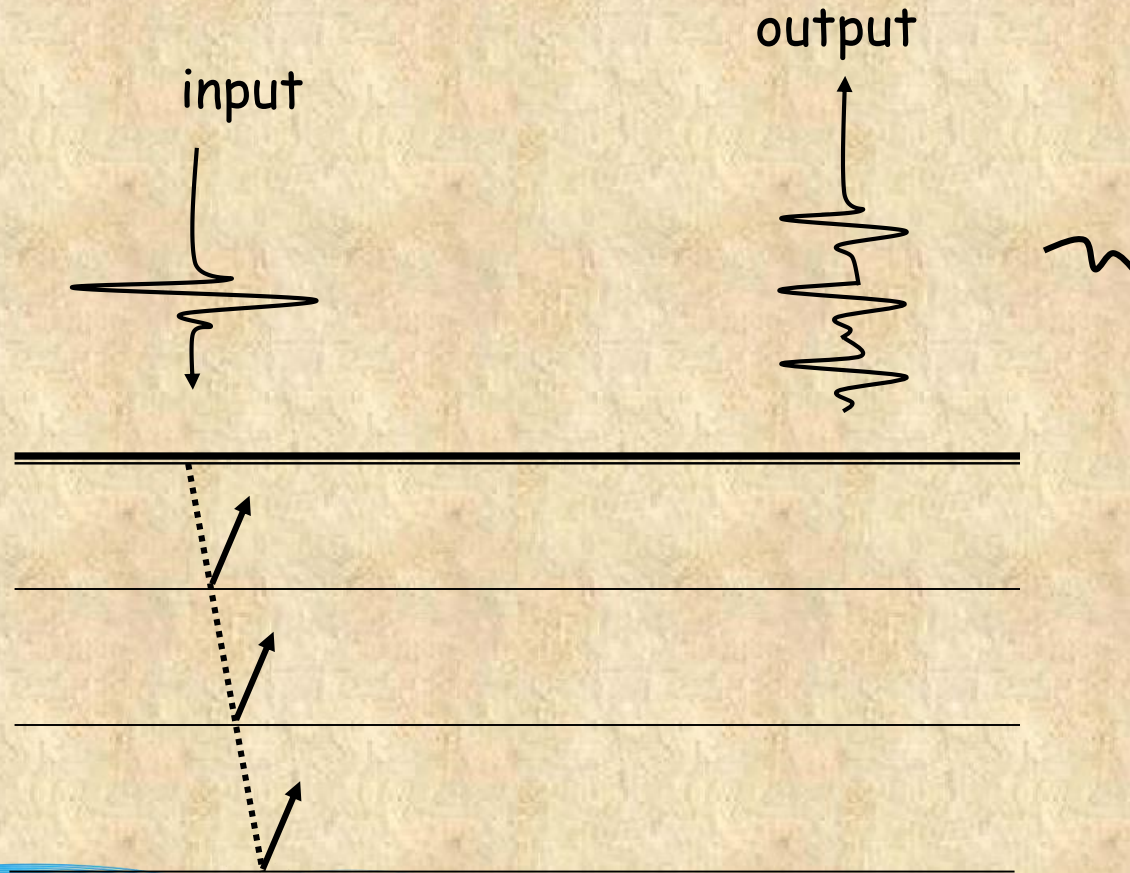
E.g., $A = [0.25 \ 0.5 \ -0.25 \ 0.75]$; $B = [1 \ 2 \ -0.5]$;

$$C = [0.2500 \ 1.0000 \ 0.6250 \ 0 \ 1.6250 \ -0.3750]$$

$$A = 0.25z^0 + 0.5z^1 - 0.25z^2 + 0.75z^3$$

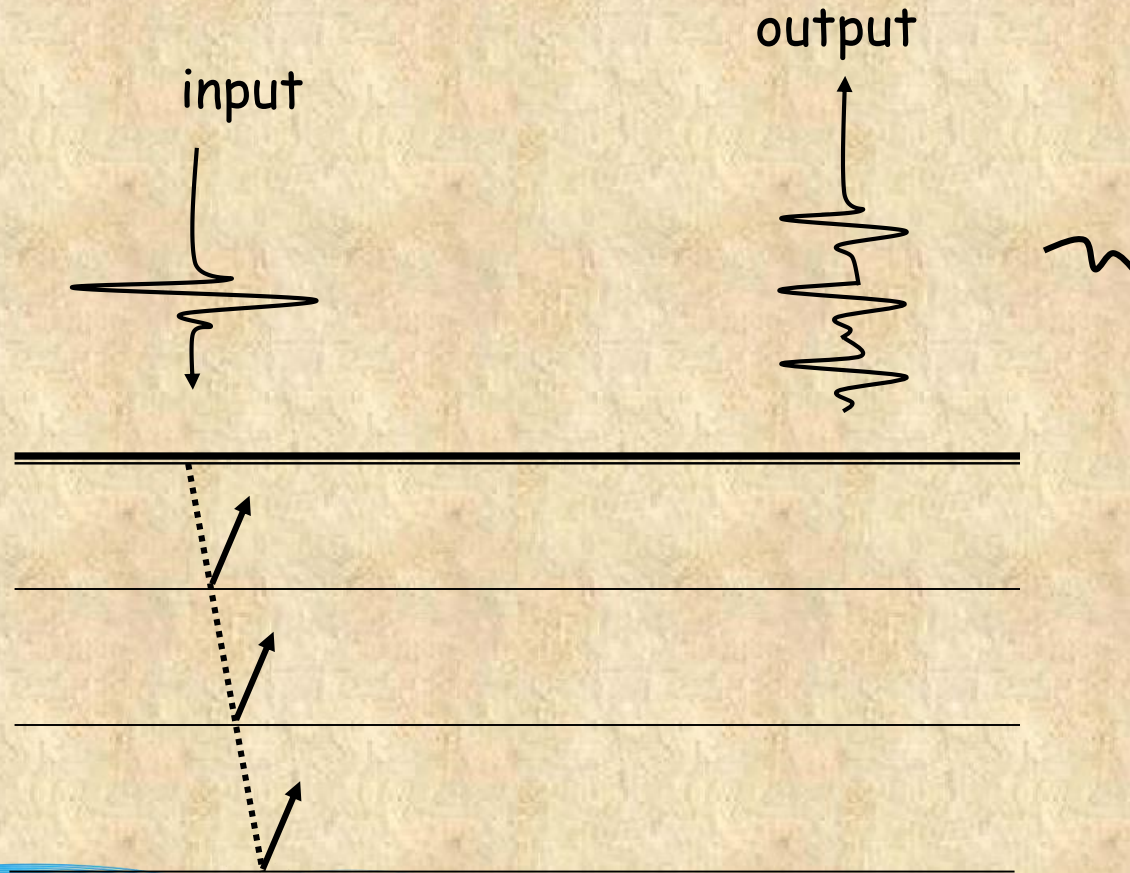
$$B = z^0 + 2z^1 - 0.5z^2$$

Convolutional Model for the Earth



Reflections in the earth are viewed as equivalent to a convolution process between the earth and the input seismic wavelet.

Convolutional Model for the Earth



$$\text{SOURCE} * \text{Reflection Coefficient} = \text{DATA}$$

(input)

(earth)

(output)

where * stands for convolution



Convolutional Model for the Earth

$$\begin{array}{ccccc} \text{SOURCE} & * & \text{Reflection Coefficient} & = & \text{DATA} \\ \text{(input)} & & \text{(earth)} & & \text{(output)} \end{array}$$

where * stands for convolution

(MORE REALISTIC)

$$\begin{array}{ccccc} \text{SOURCE} & * & \text{Reflection Coefficient} & + \text{noise} & = \text{DATA} \\ \text{(input)} & & \text{(earth)} & & \text{(output)} \end{array}$$

$$s(t) * e(t) + n(t) = d(t)$$

Convolution in the TIME domain is equivalent to MULTIPLICATION in the FREQUENCY domain

$$\begin{array}{ccccccc} s(t) & * & e(t) & + n(t) & = & d(t) \\ \downarrow \text{FFT} & & \downarrow \text{FFT} & \downarrow \text{FFT} & & \\ s(f, \text{phase}) \times e(f, \text{phase}) + n(f, \text{phase}) & & & & = & d(f, \text{phase}) \\ & & & & \downarrow \text{Inverse FFT} & \\ & & & & d(t) & \end{array}$$

Convolution

$$(f * g)(t) = \int_{-\infty}^{\infty} f(t - \tau) g(\tau) d\tau.$$



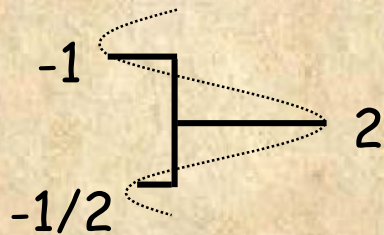
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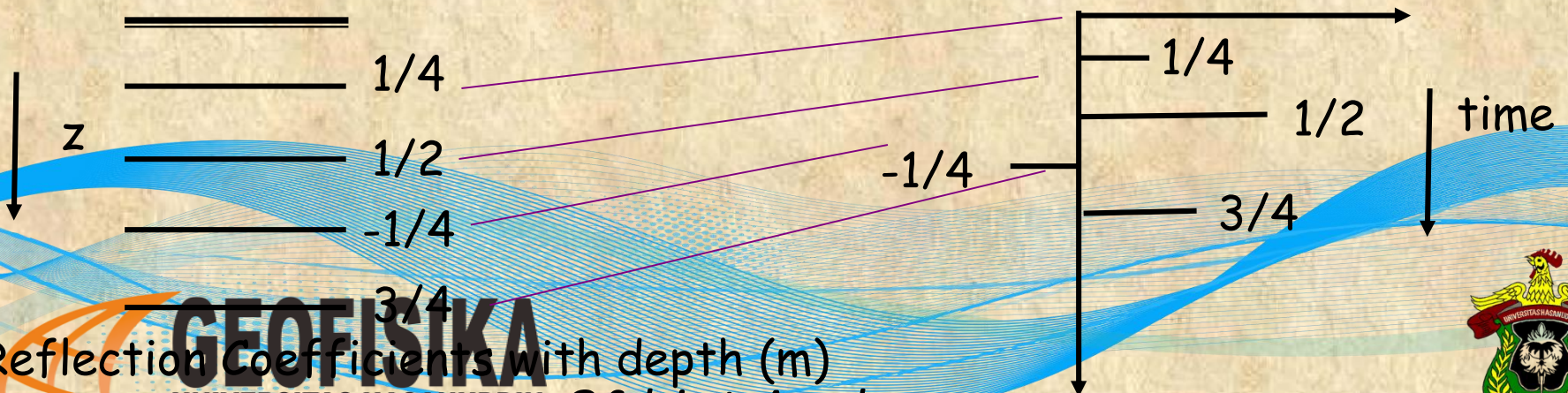
CONVOLUTION as a mathematical operator

signal has 3 terms ($j=3$)



$$D_j = \sum_{k=0}^j s_{j-k} e_k$$

earth has 4 terms ($k=4$)



$\downarrow$
 0
 $-1/2$
 2
 $0 \times 1 = 0$
 $0 \times 0 = 0$
 $0 \times 0 = 0 +$
 $1/4 \times 0 = 0$
 $1/2 \times 0 = \frac{0}{0}$
 $-1/4$
 $3/4$
 0
 0
 0





0

0

-1/2

$$0 \times 2 = 0$$

$$0 \times -1 = 0$$

$$0 \times 0 = 0 +$$

$$1/4 \times 0 = 0$$

$$1/2 \times 0 = 0$$

$$-1/4 \quad 0 = \frac{0}{0}$$

3/4

0

0

0



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$$\begin{array}{rcl} & 0 & \\ & 0 & \\ & 0 & \\ \downarrow & 0 \times -1/2 & = 0 \\ & 0 \times 2 & = 0 \\ & 0 \times 1 & = 0 + \\ 1/4 \times & 0 & = 0 \\ 1/2 \times & 0 & = 0 \\ -1/4 \times & 0 & = 0 \\ 3/4 \times & 0 & = 0 \\ & 0 & \\ & 0 & \\ & 0 & \end{array}$$



$$\begin{array}{rcl}
 & 0 & \\
 & 0 & \\
 \downarrow & 0 \times 0 & = 0 \\
 & 0 \times -1/2 & = 0 \\
 & 0 \times 2 & = 0 + \\
 & 1/4 \times 1 & = 1/4 \\
 & 1/2 \times 0 & = 0 \\
 & -1/4 \times 0 & = 0 \\
 & 3/4 \times 0 & = 0 \\
 & 0 \times 0 & = 0 \\
 & 0 & \underline{1/4} \\
 & 0 &
 \end{array}$$



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$$\begin{array}{rcl}
 & 0 & \\
 0 \times 0 & = & 0 \\
 0 \times 0 & = & 0 \\
 0 \times -1/2 & = & 0 + \\
 1/4 \times 2 & = & 1/2 \\
 1/2 \times 1 & = & 1/2 \\
 -1/4 \times 0 & = & 0 \\
 3/4 \times 0 & = & 0 \\
 0 \times 0 & = & 0 \\
 0 \times 0 & = & 0 \\
 \hline
 0 & & 1
 \end{array}$$



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$$0 \times 0 = 0$$

$$0 \times 0 = 0$$

$$0 \times 0 = 0 +$$

$$1/4 \times -1/2 = -1/8$$

$$1/2 \times 2 = 1$$

$$-1/4 \times 1 = -1/4$$

$$3/4 \times 0 = 0$$

$$0 \times 0 = 0$$

$$0 \times 0 = 0$$

$$0 \times 0 = 0$$

$$\underline{\hspace{1cm}} \\ 5/8$$



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$$0 \times = 0$$

$$0 \times 0 = 0$$

$$0 \times 0 = 0 +$$

$$1/4 \times 0 = 0$$

$$1/2 \times -1/2 = -1/4$$

$$-1/4 \times 2 = -1/2$$

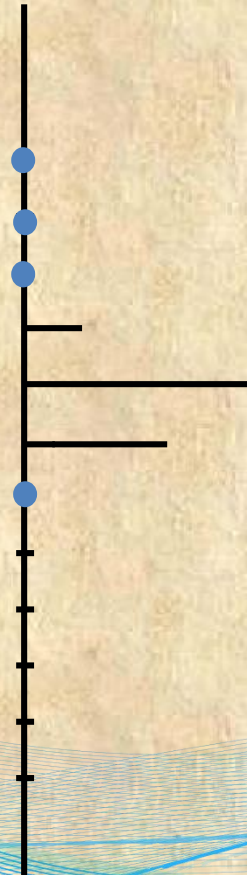
$$3/4 \times 1 = 3/4$$

$$0 \times 0 = 0$$

$$0 \times 0 = 0$$

$$0 \times 0 = 0$$

$$0$$



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0

0

$$0 \times 0 = 0 +$$

$$1/4 \times 0 = 0$$

$$1/2 \times 0 = 0$$

$$-1/4 \times -1/2 = 1/8$$

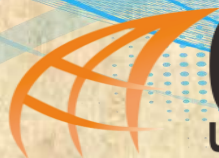
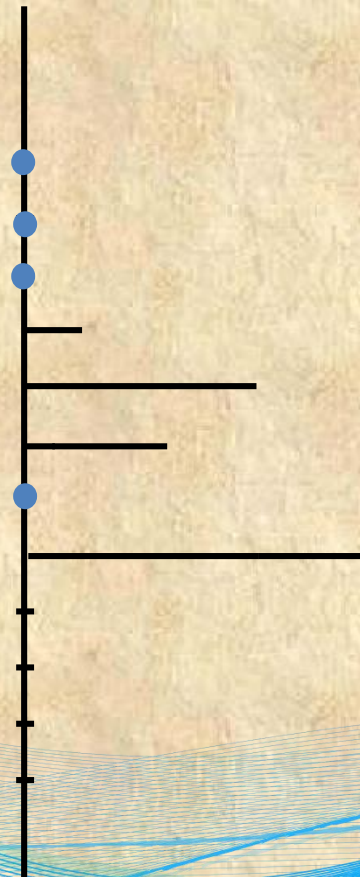
$$3/4 \times 2 = 1 \frac{1}{2}$$

$$0 \times 1 = 0$$

$$0 \times 0 = 0$$

$$0 \times 0 = 0$$

$$0 \times 0 = 1 \frac{5}{8}$$



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0

0

0

+

$$1/4 \times 0 = 0$$

$$1/2 \times 0 = 0$$

$$-1/4 \times 0 = 0$$

$$3/4 \times -1/2 = -3/8$$

$$0 \times 2 = 0$$

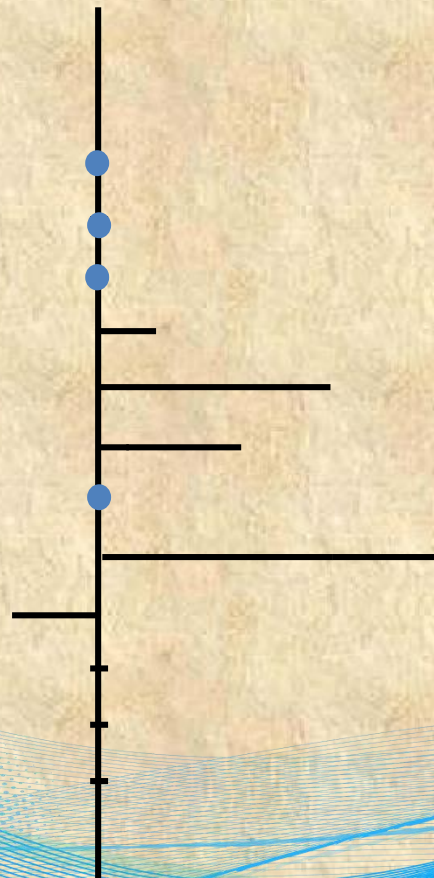
$$0 \times 1 = 0$$

$$0 \times 0 = 0$$

0

-3/8

0



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0

0

0

1/4

$$1/2 \times 0 = 0$$

$$-1/4 \times 0 = 0$$

$$3/4 \times 0 = 0$$

$$0 \times -1 = 0$$

$$0 \times 2 = 0$$

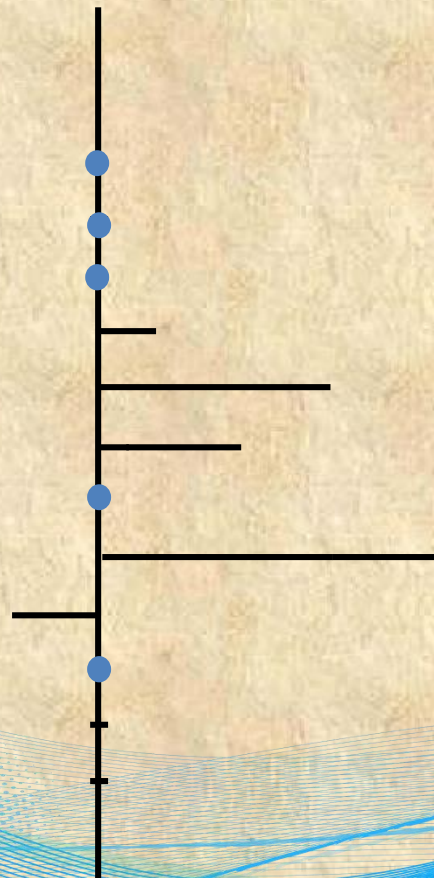
$$0 \times -1/2 = 0$$

0

0

0

+



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MATLAB

%convolution

a = [0.25 0.5 -0.25 0.75]; b = [1 2 -0.5];

c = conv(a,b)

d = deconv(c,a)

$$a = 0.25z + 0.5z^2 - 0.25z^3 + 0.75z^4$$

$$b = z + 2z^2 - 5z^3$$

c = 0.2500 1.0000 0.6250 0 1.6250 -0.3750

matlab

Spiking Deconvolution

In order to compress seismic signal in time and whiten the spectrum.

Advantages: shows embedded signal in noise

Disadvantages: heightens noise



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Convolutional model

$$data(t) = earth(t) * wavelet(t)$$

$$data(\nu, \phi) = earth(\nu, \phi) \times wavelet(\nu, \phi)$$

Steps in Spiking Deconvolution

- Calculate autocorrelation function (ACF)
- Estimate second crossing of ACF in s
- Conduct inverse filtering using ACF

Significant arrivals in SP 2

Calama, Chile

(181207_1)

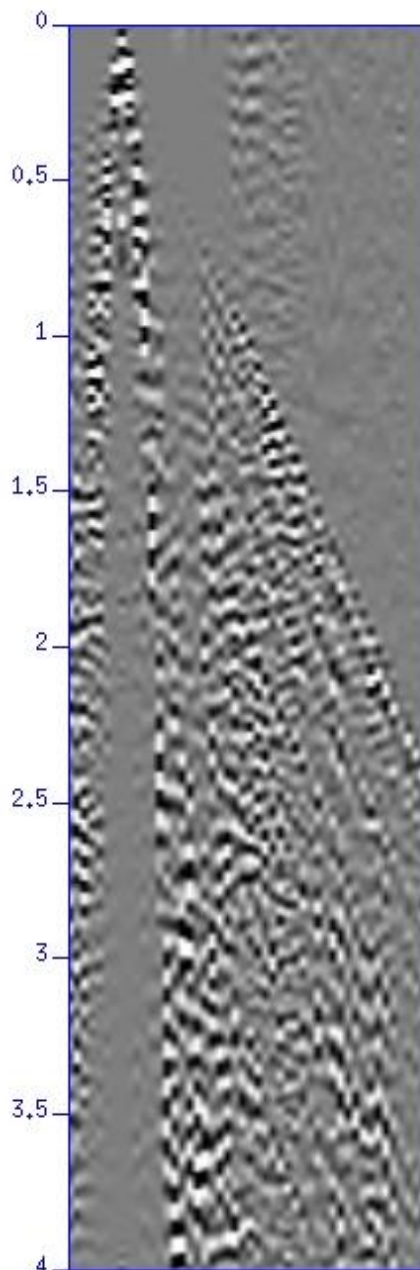
- Refractions (mainly) and reflections

X Trace look -10000_10000

W

offset (m)

E

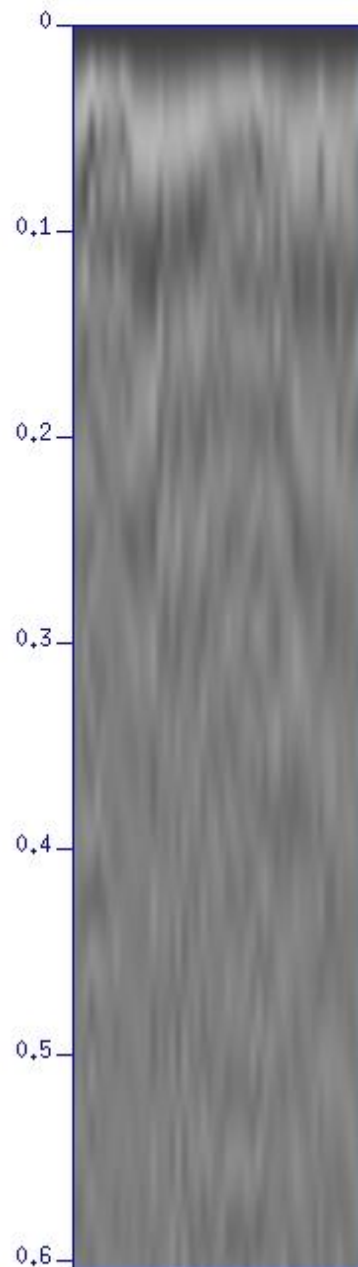


TWTT (s)

Z 181207 1 f=0,5,20,25 pbal

X Autocorrelation <3>

offset (m)

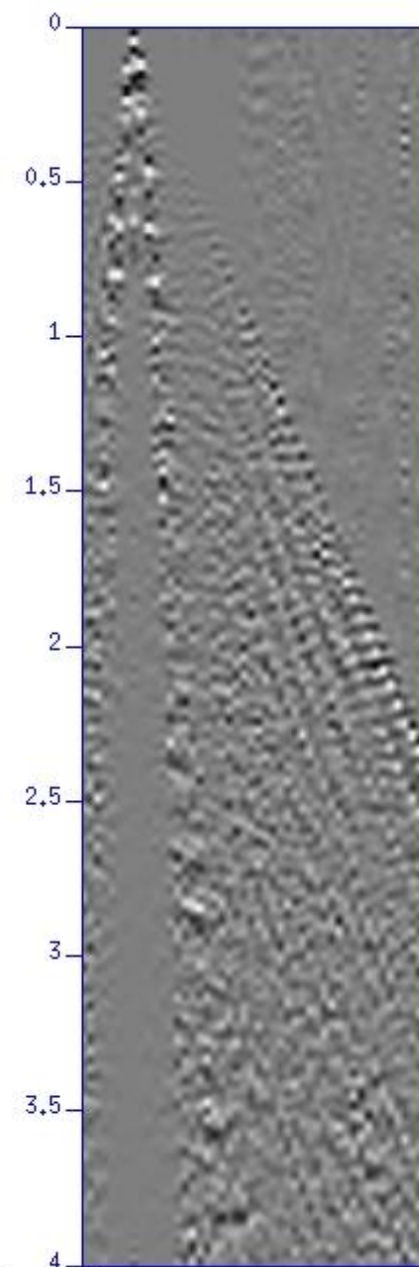


Time(s): ns x dt

Z 181207 1 303 samples

X Spiking Deconvolution <2>

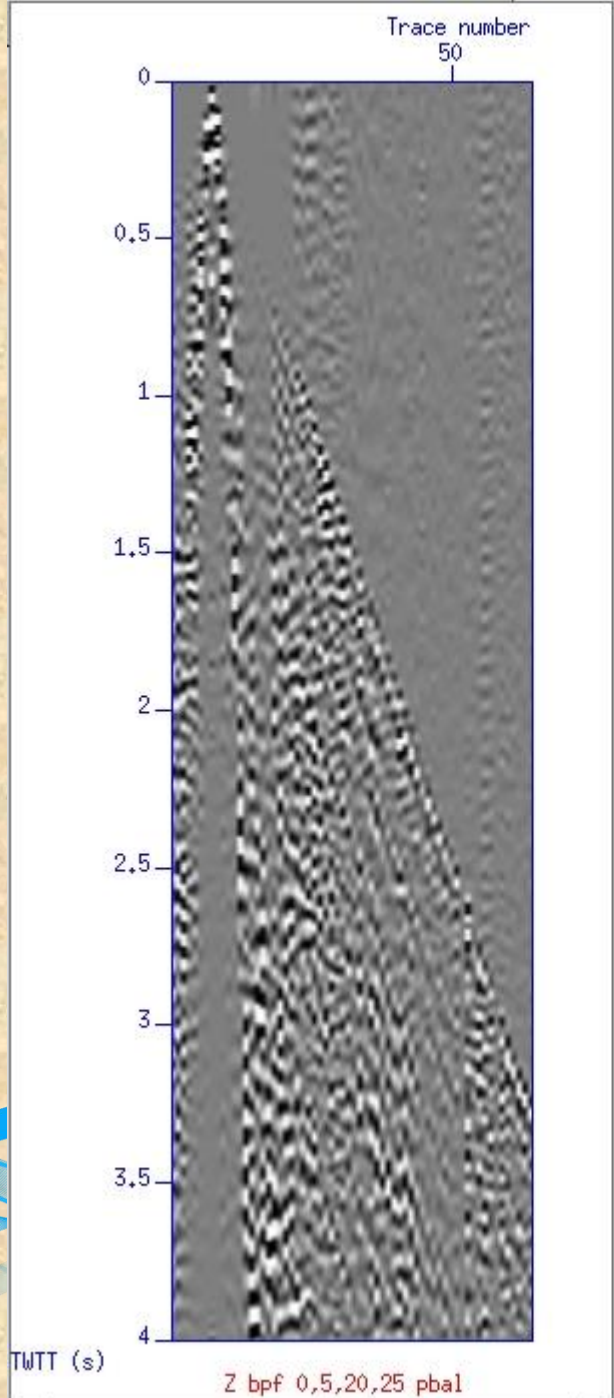
offset (m)



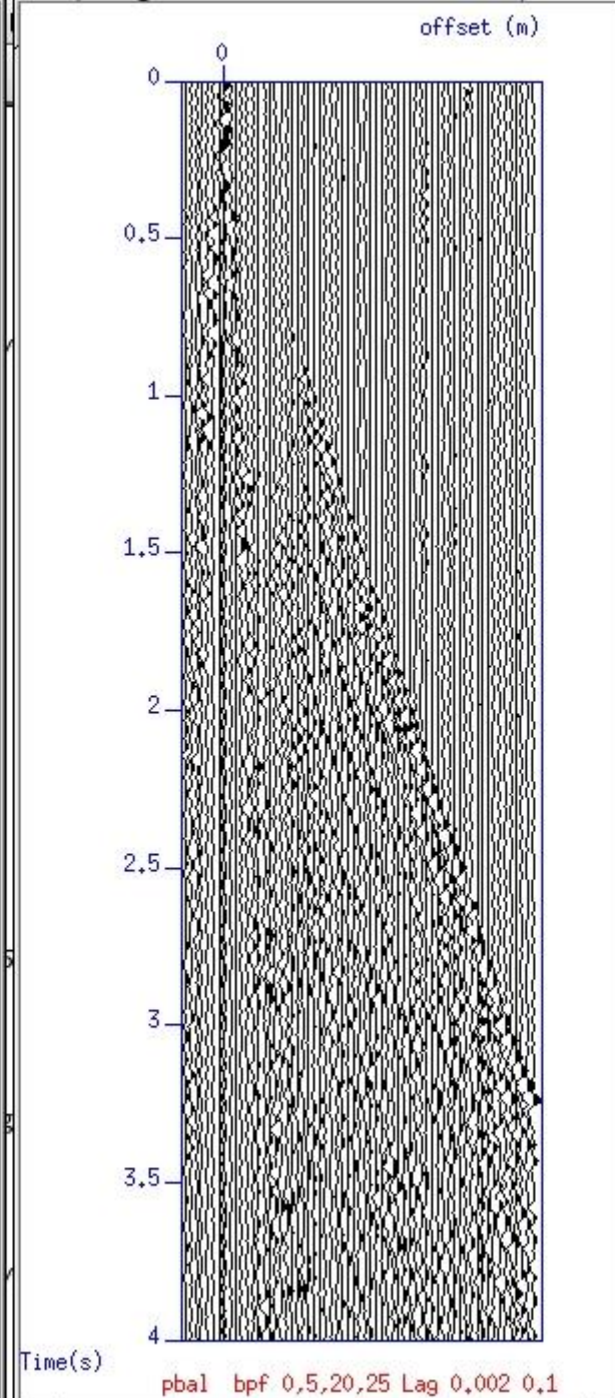
Time(s): ns x dt

Z 181207 1 Lag 0,002,0,1 s

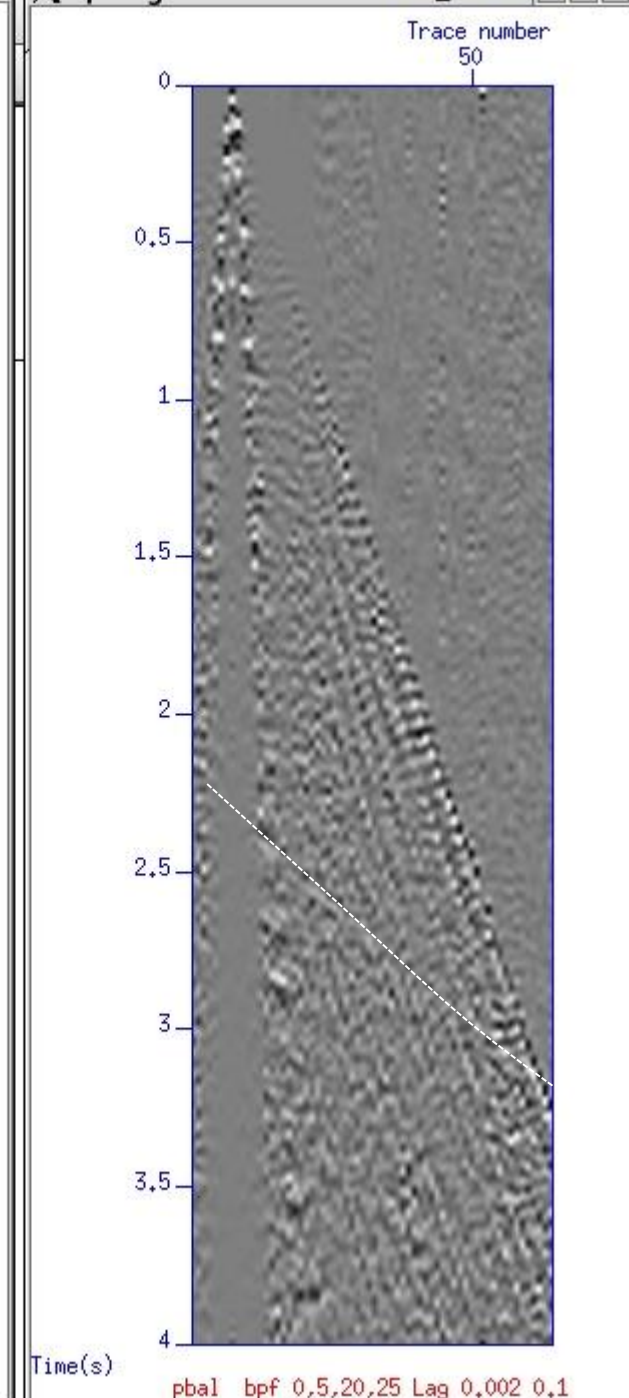
X Trace look -20000_20000 181207_1



X Spiking Deconvolution181207_1 <2>



X Spiking Deconvolution181207_1

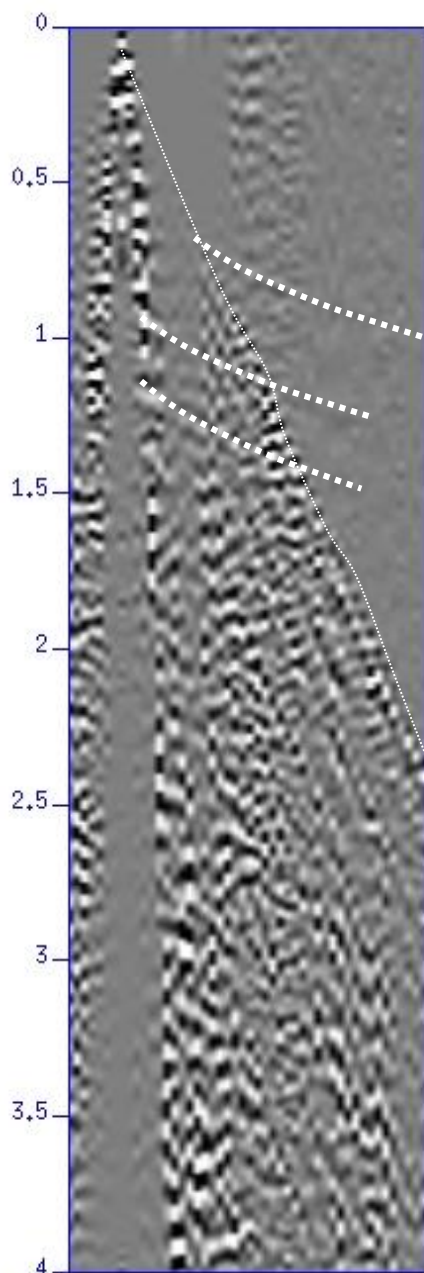


Trace look -10000_10000

W

offset (m)

E

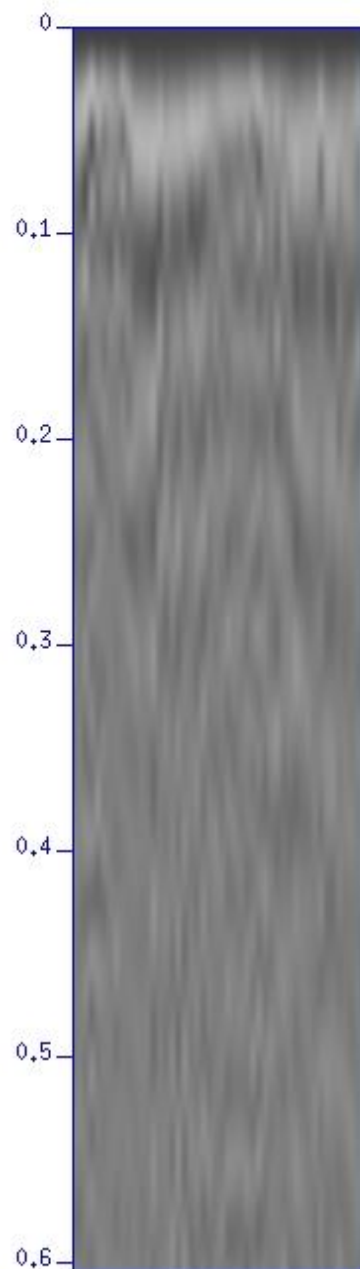


TWTT (s)

Z 181207 1 f=0,5,20,25 pbal

Autocorrelation <3>

offset (m)

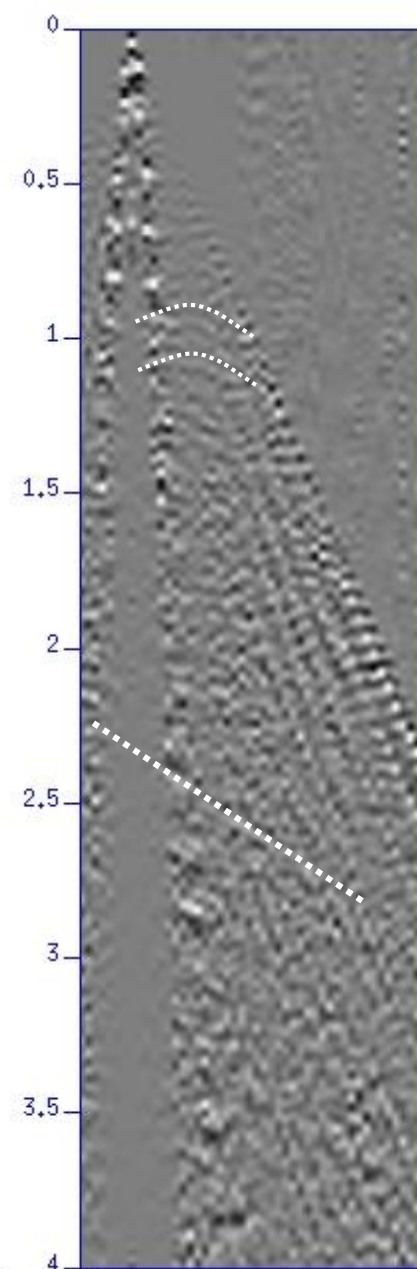


Time(s): ns x dt

Z 181207 1 303 samples

Spiking Deconvolution <2>

offset (m)



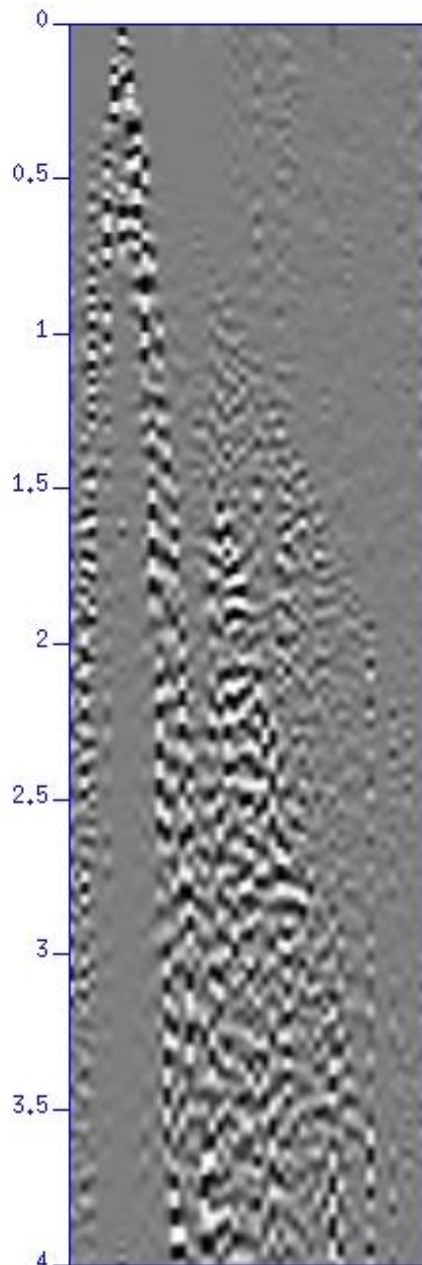
Time(s): ns x dt

Z 181207 1 Lag 0,002,0,1 s

X Trace look -10000_10000 <2>

offset (m)

W



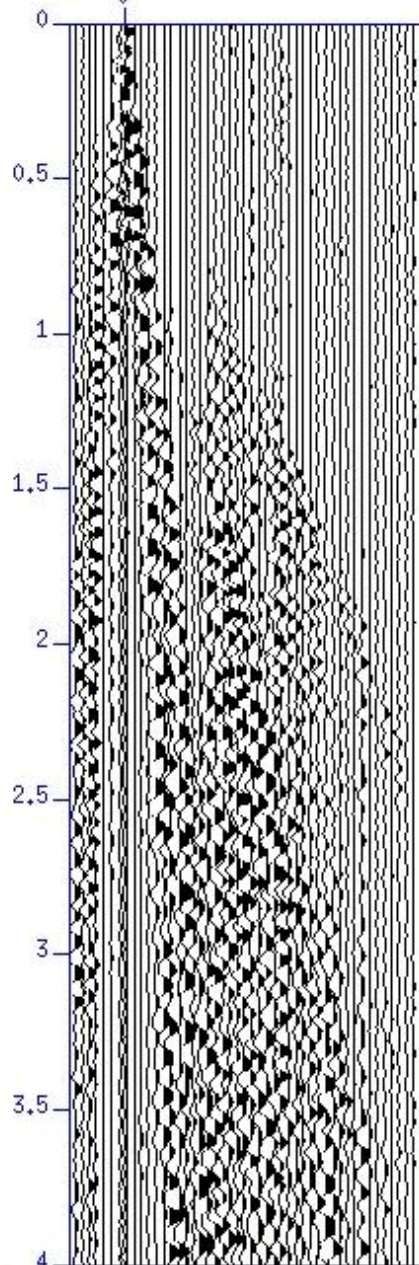
TwTT (s)

Radial component 181207 1 f=0,5,20,25 pbal

X Trace look -10000_10000

offset (m)

E

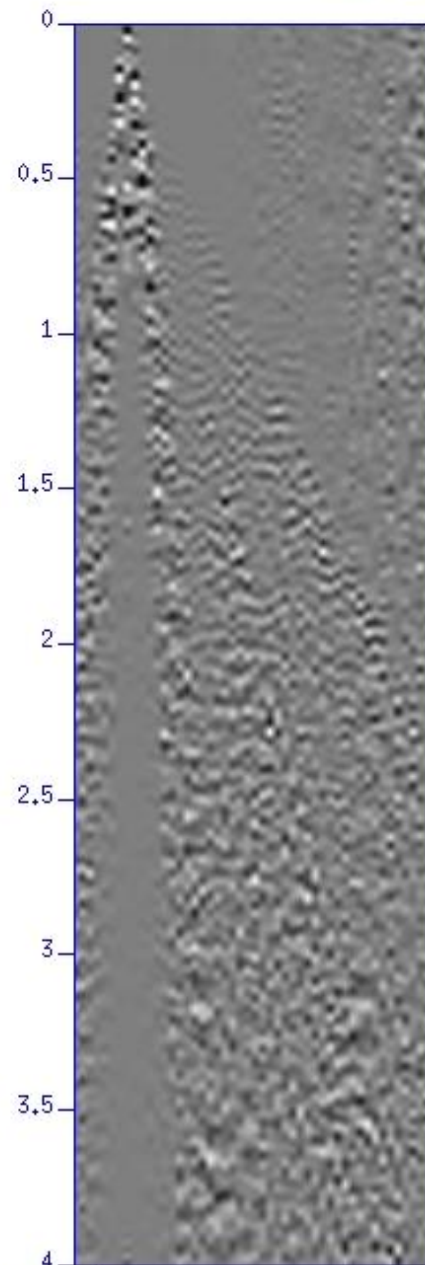


TwTT (s)

Radial component 181207 1 f=0,5,20,25 pbal

X Spiking Deconvolution <3>

offset (m)



Time(s): ns x dt

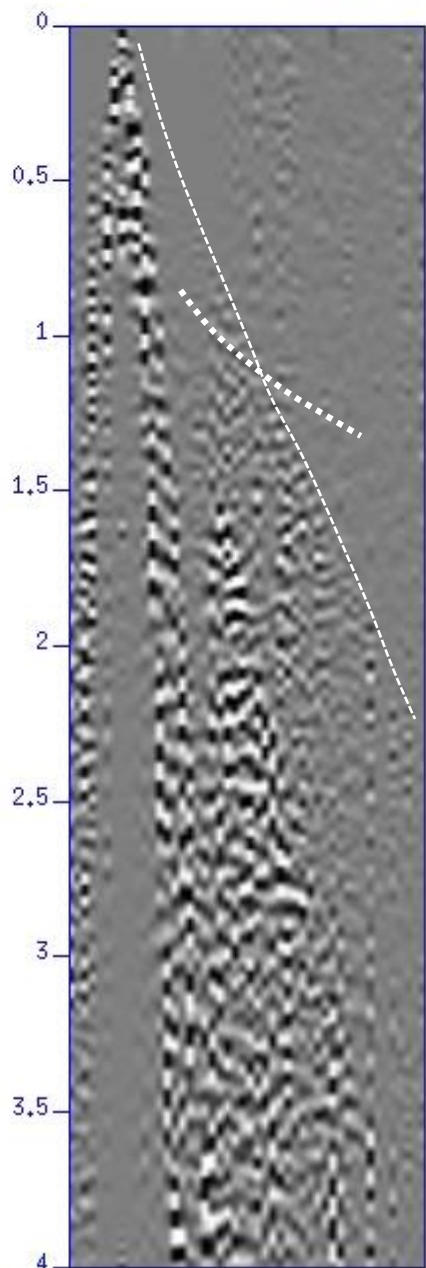
Radial component 181207 1 Lag 0,002,0,1 s

X Trace look -10000_10000 <2>

offset (m)

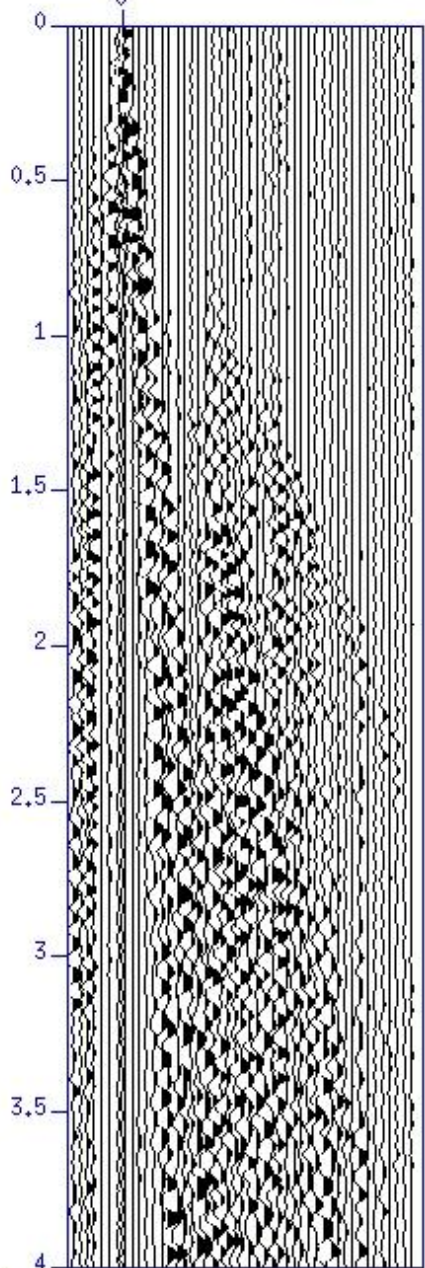
W

E



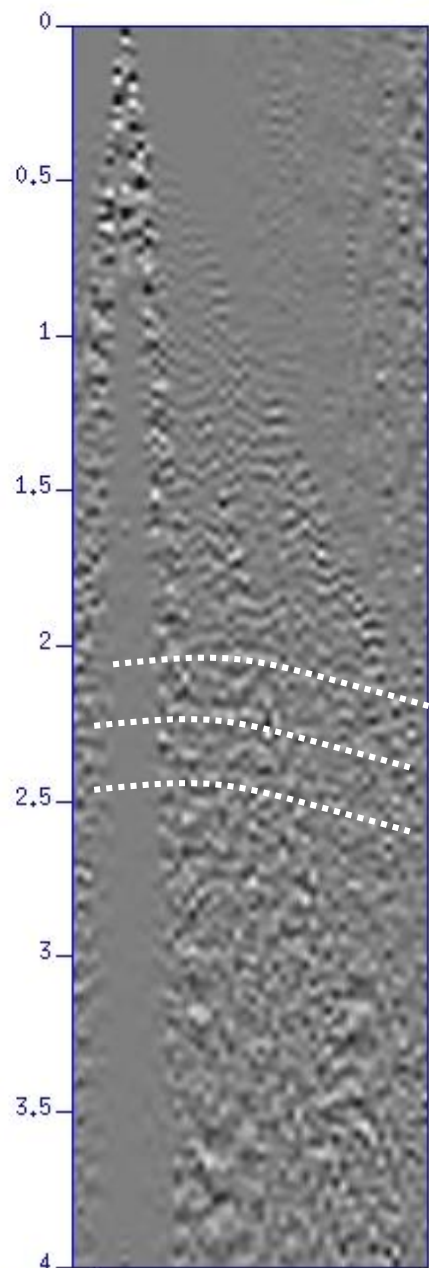
X Trace look -10000_10000

offset (m)



X Spiking Deconvolution <3>

offset (m)



TWTT (s)

Radial component 181207 1 f=0,5,20,25 pbal

TWTT (s)

Radial component 181207 1 f=0,5,20,25 pbal

Time(s): ns x dt

Radial component 181207 1 Lag 0,002,0,1 s

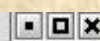
X Trace look -10000_10000



X Spiking Deconvolution <2>



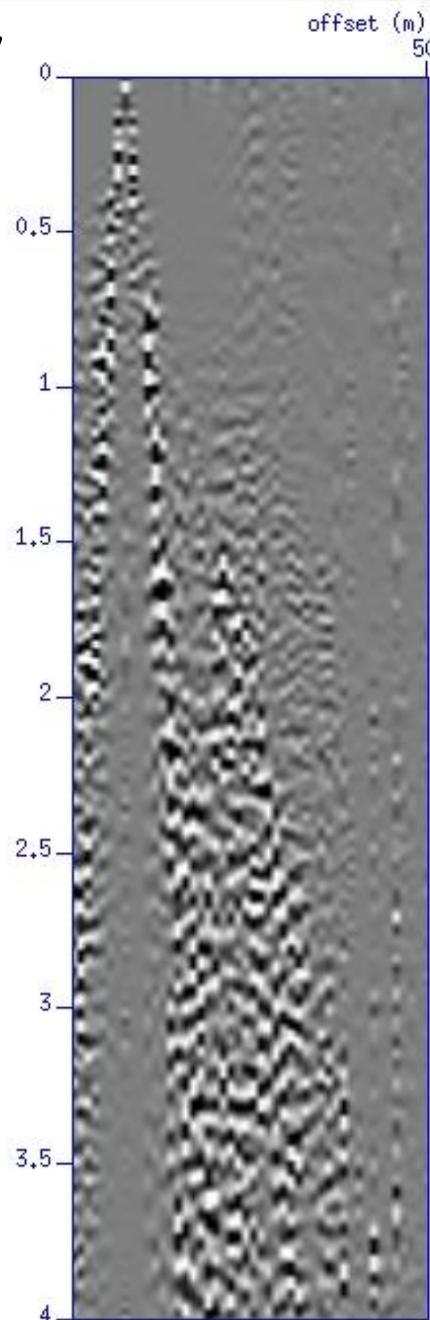
X Spiking Deconvolution



W

offset (m)

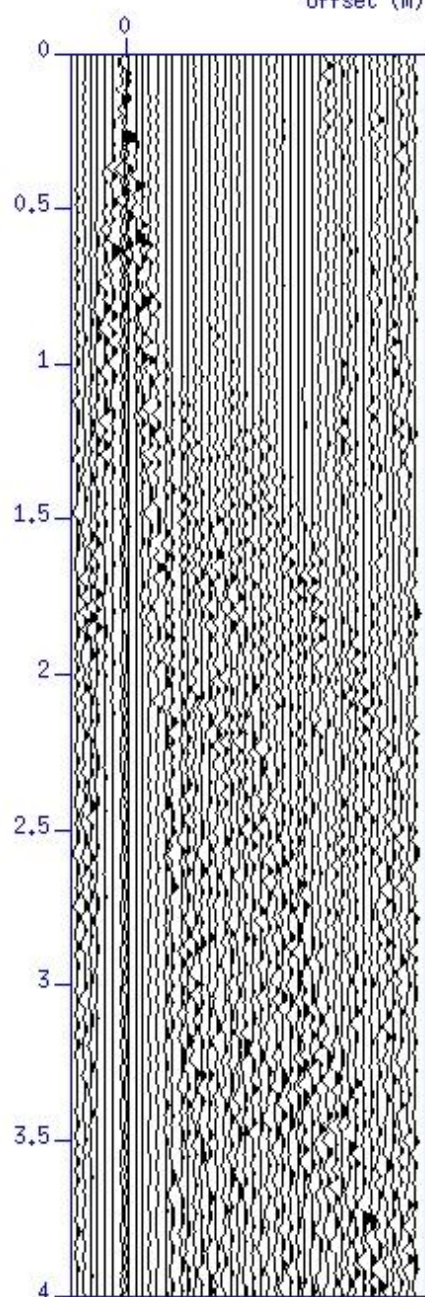
E



Time(s)

Ht 181207 1 f=0,5,20,25 pbal

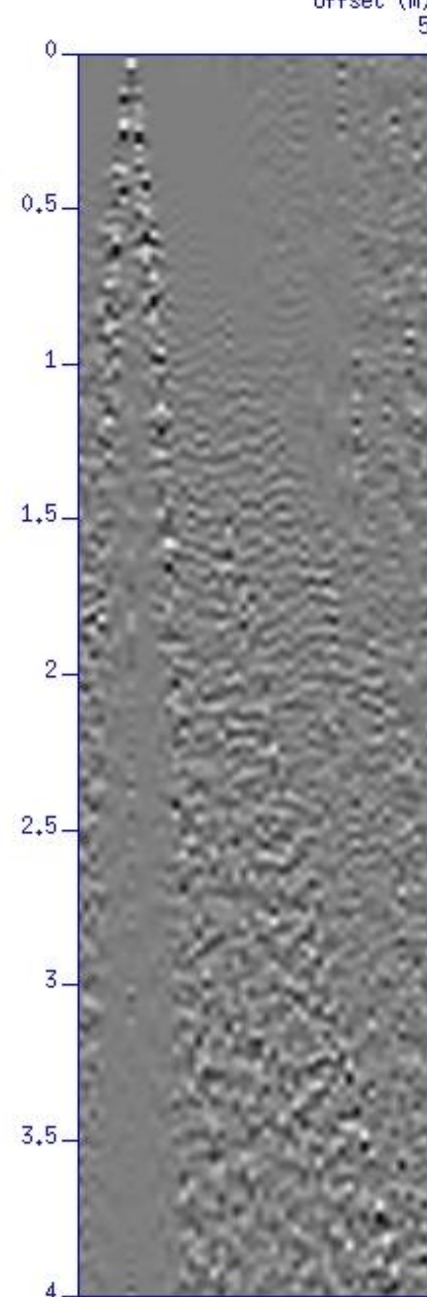
offset (m)



Time(s): ns x dt

Ht 181207 1 Lag 0.002,0.1 s

offset (m)



Time(s): ns x dt

Ht 181207 1 Lag 0.002,0.1 s

X Trace look -10000_10000



X Spiking Deconvolution <2>



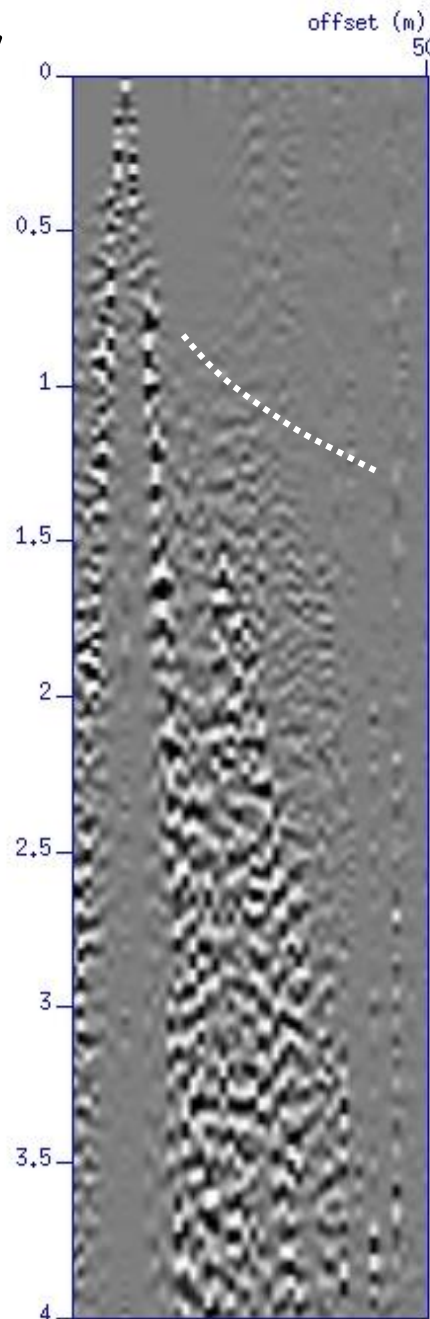
X Spiking Deconvolution



W

offset (m)

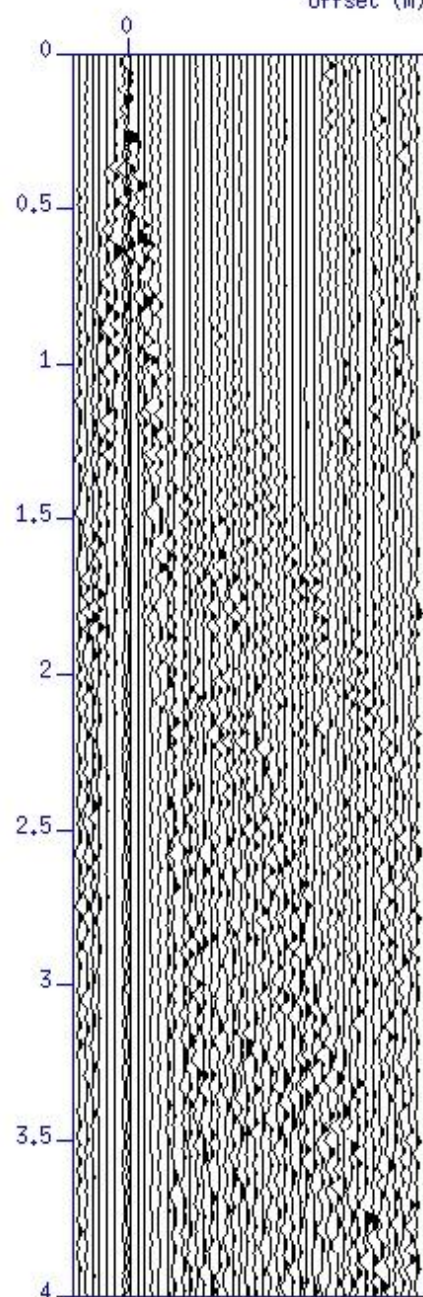
E



Time(s)

Ht 181207 1 f=0,5,20,25 pbal

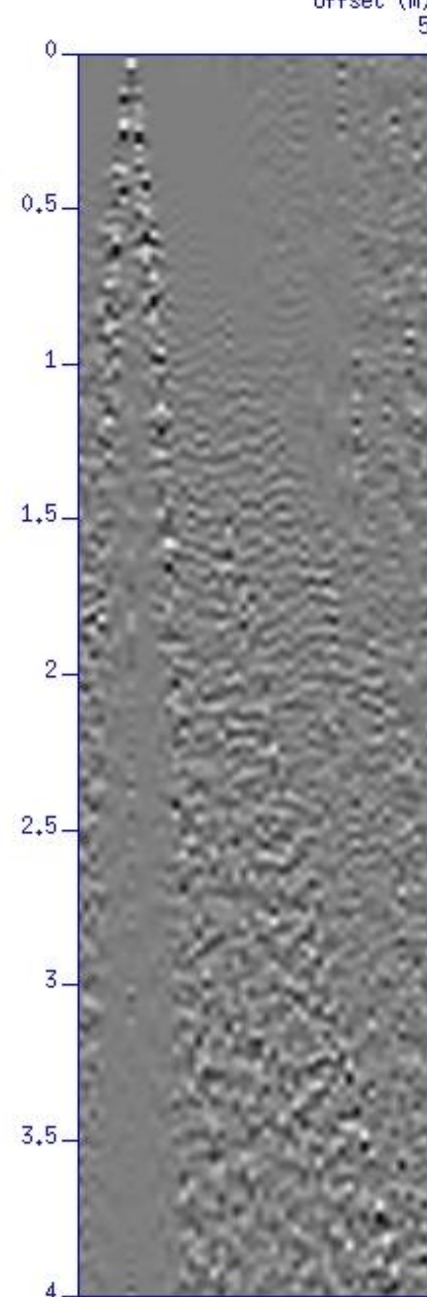
offset (m)



Time(s): ns x dt

Ht 181207 1 Lag 0,002,0,1 s

offset (m)



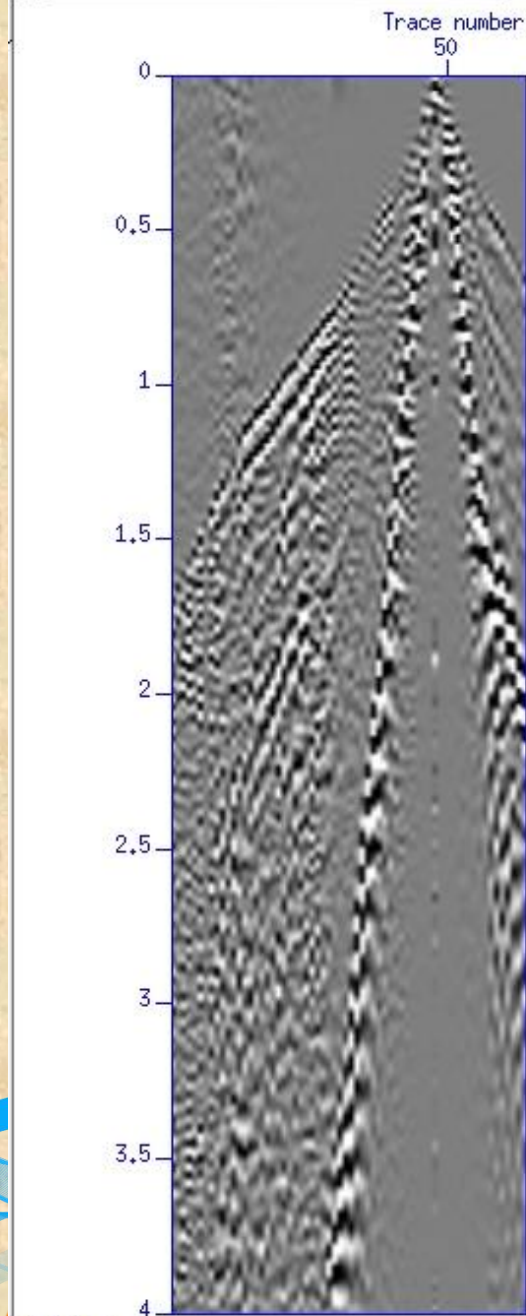
Time(s): ns x dt

Ht 181207 1 Lag 0,002,0,1 s

Significant arrivals in SP 5 (191207_1)

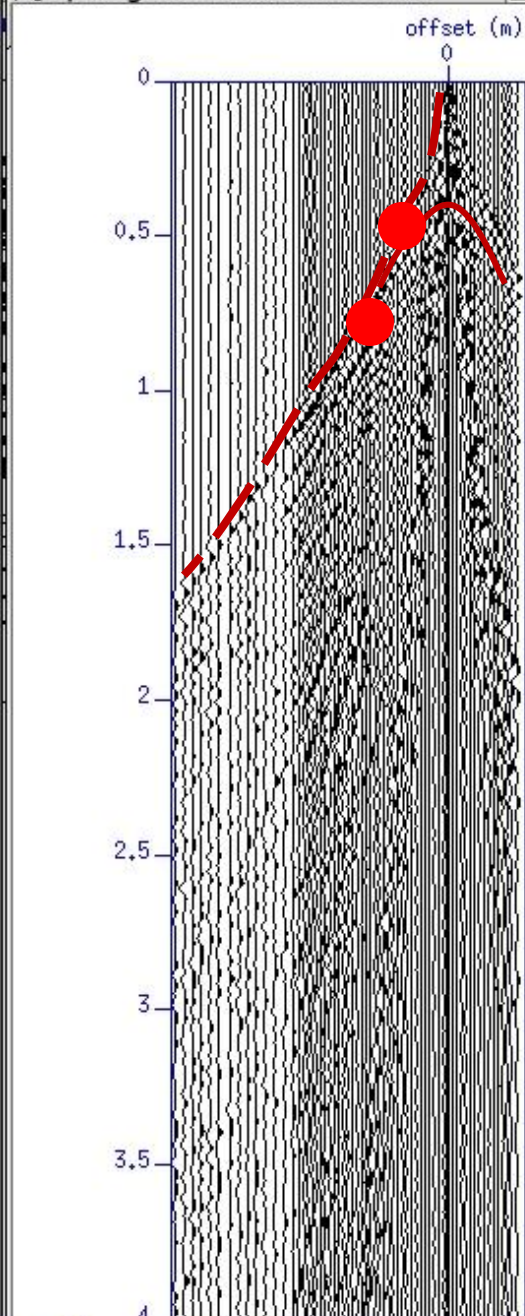
- Refractions (mainly)

X Trace look -10000_10000 191207_1



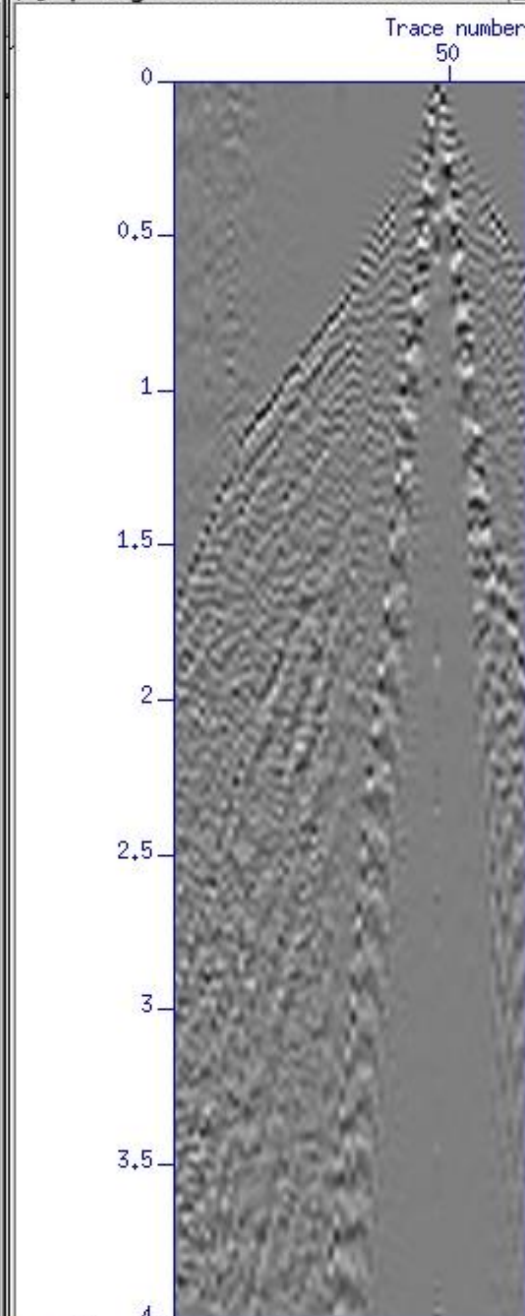
Z bpf 0,5,20,25 pbal

X Spiking Deconvolution191207_1



pbal bpf 0,5,20,25 Lag 0.002 0,1

X Spiking Deconvolution191207_1 <3>

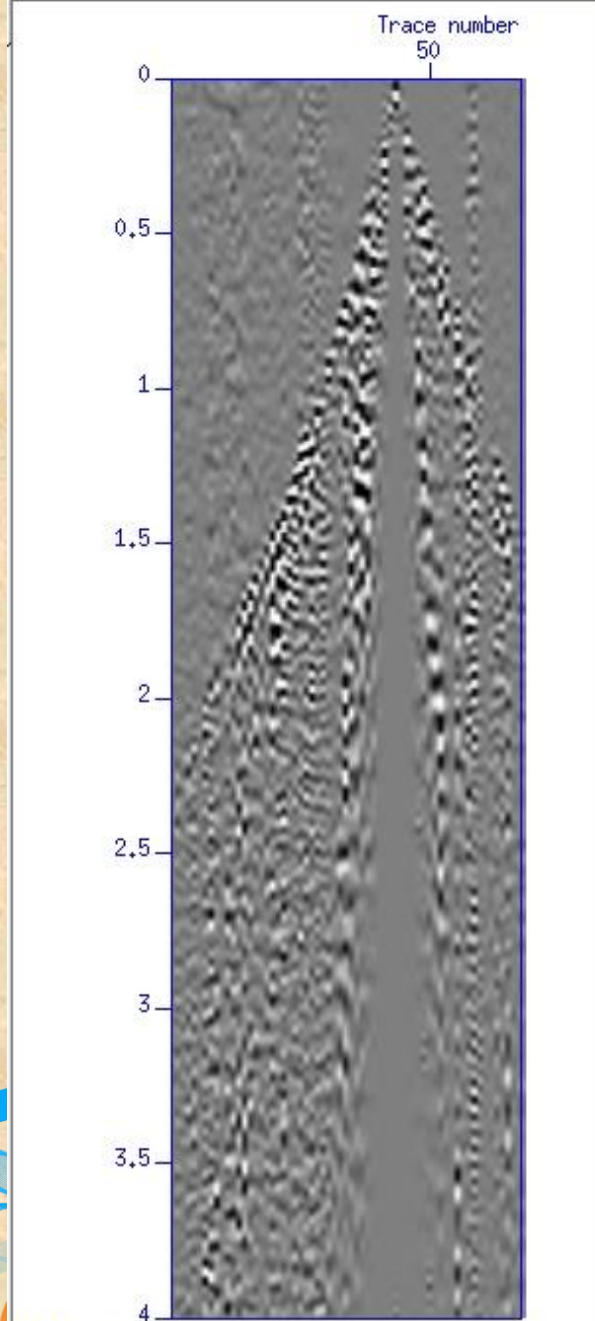


pbal bpf 0,5,20,25 Lag 0.002 0,1

Significant arrivals in SP 6 (181207_3)

- Refractions (mainly)

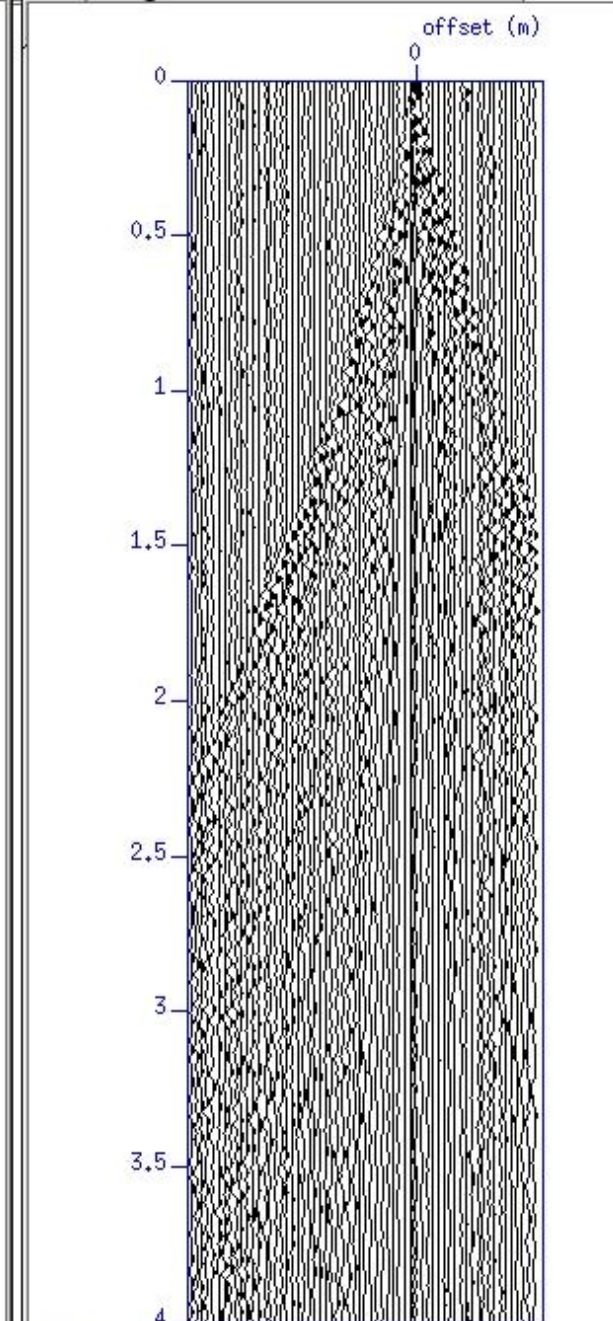
X Trace look -10000_10000 181207_3



TWTT (s)

Z bpf 0,5,20,25 pbal

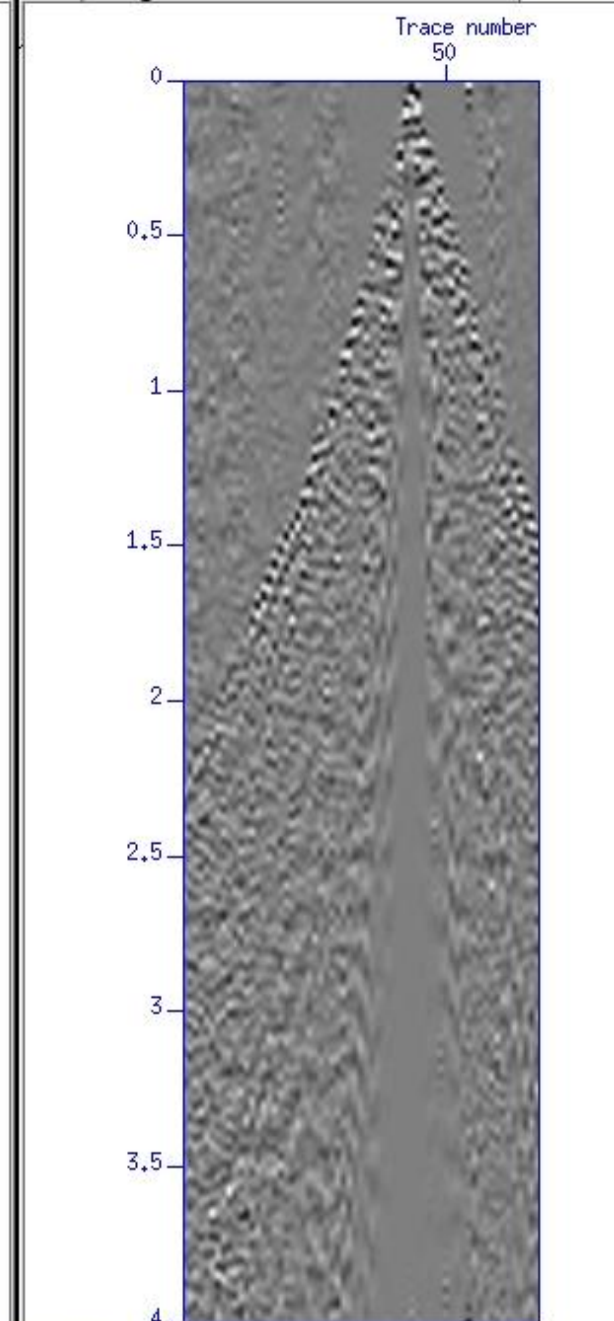
X Spiking Deconvolution181207_3 <2>



Time(s)

pbal bpf 0,5,20,25 Lag 0,002 0,1

X Spiking Deconvolution181207_3



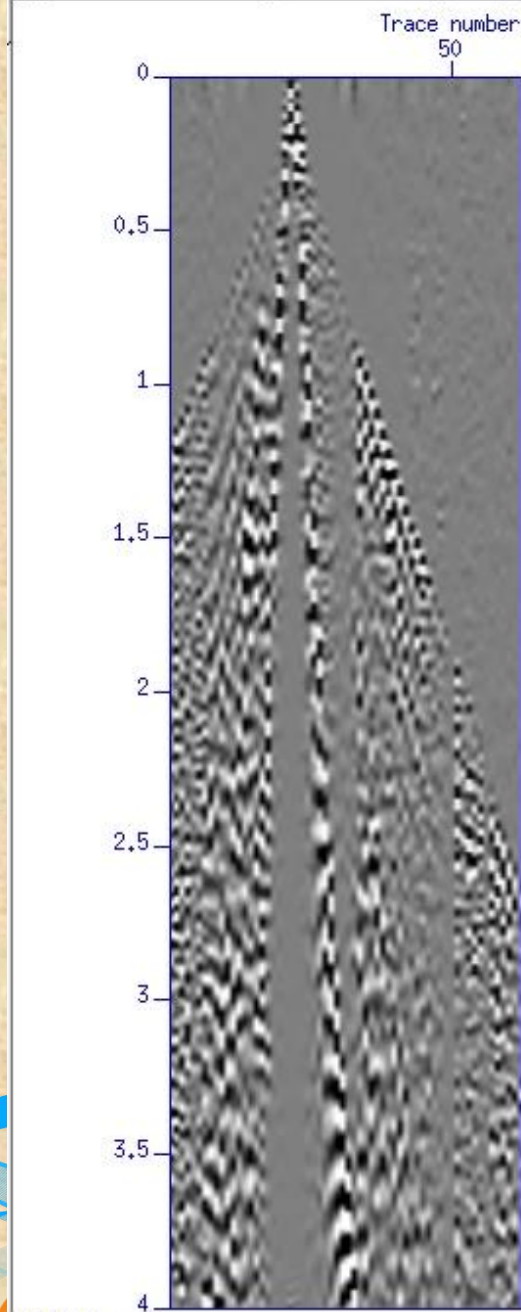
Time(s)

pbal bpf 0,5,20,25 Lag 0,002 0,1

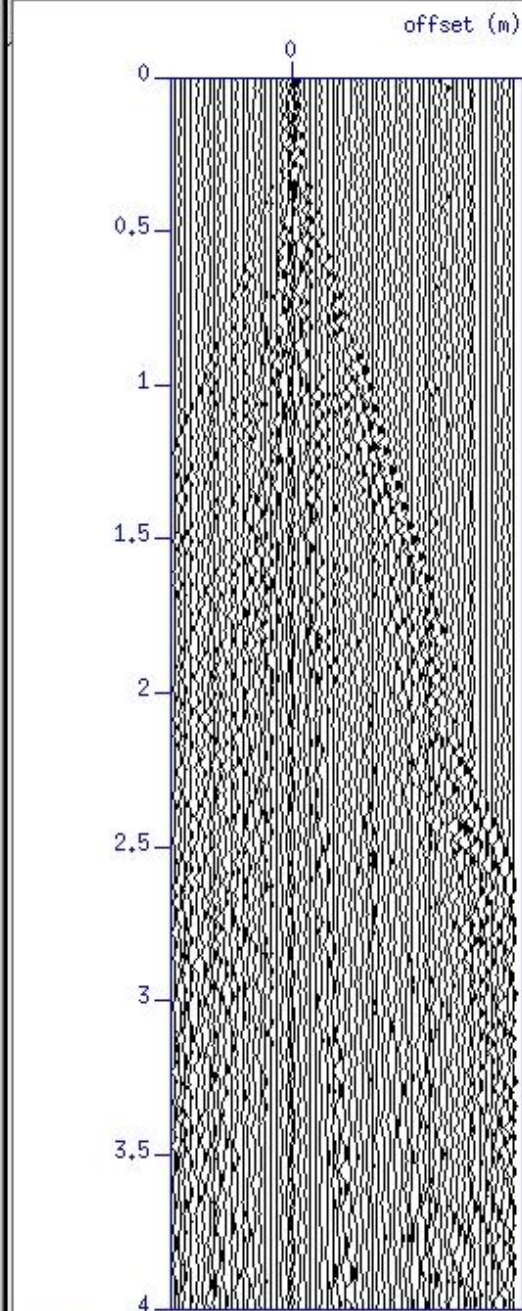
Significant arrivals in SP 4A (171207_1)

- Refractions (mainly)

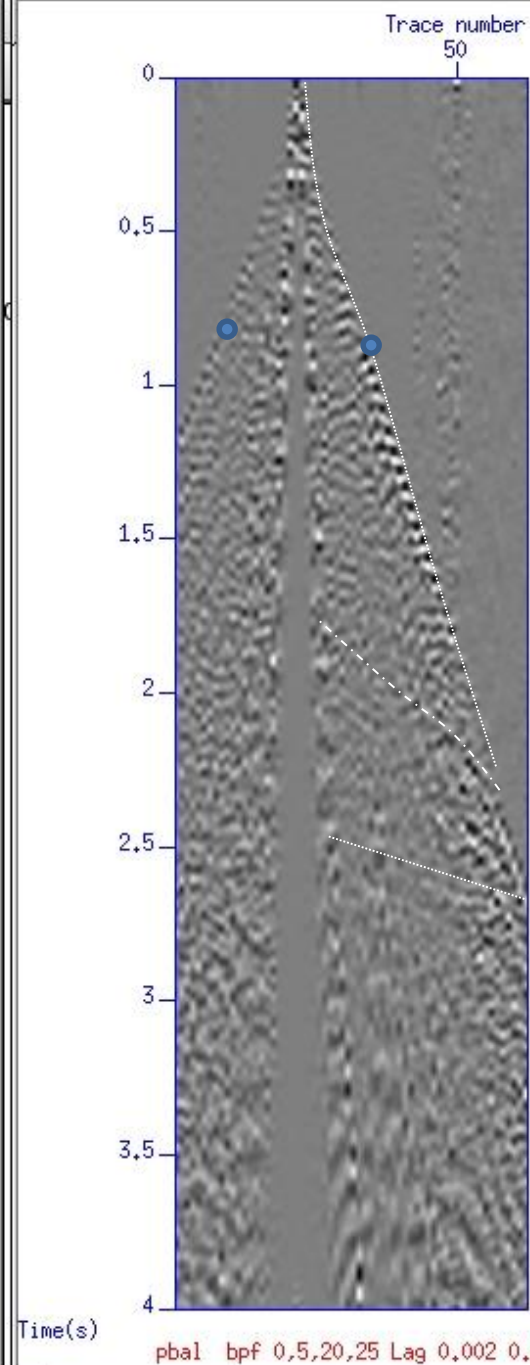
Trace look -10000_10000 171207_1



Spiking Deconvolution171207_1 <2>



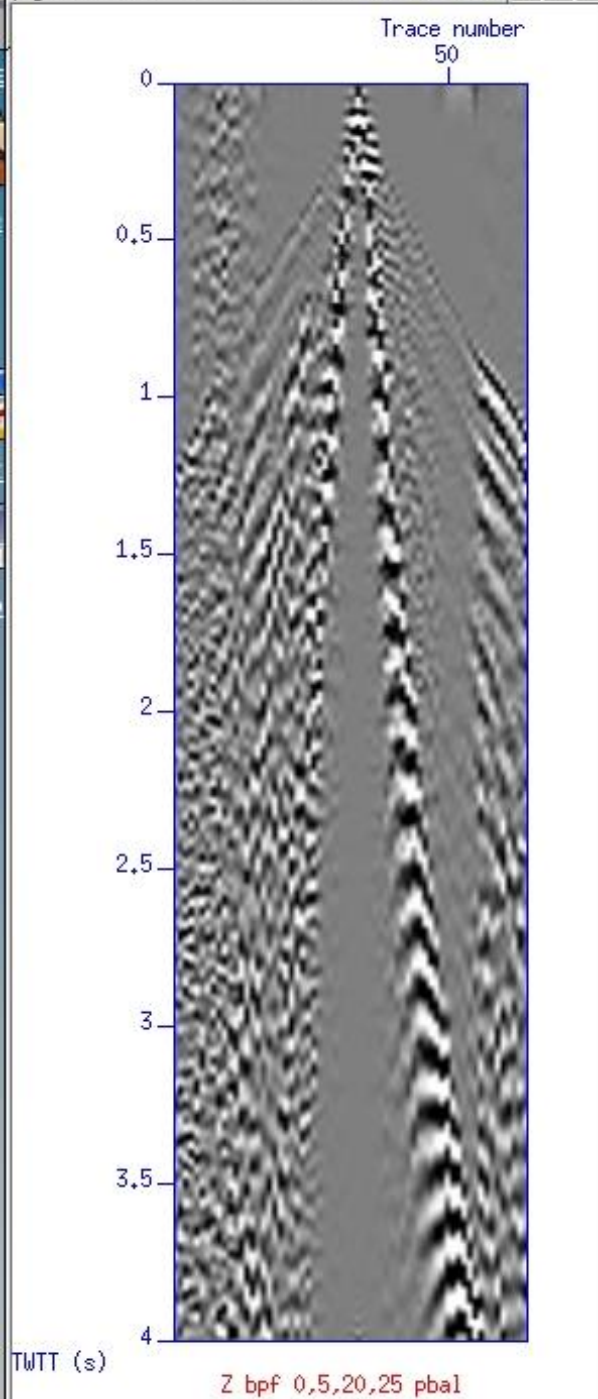
Spiking Deconvolution171207_1



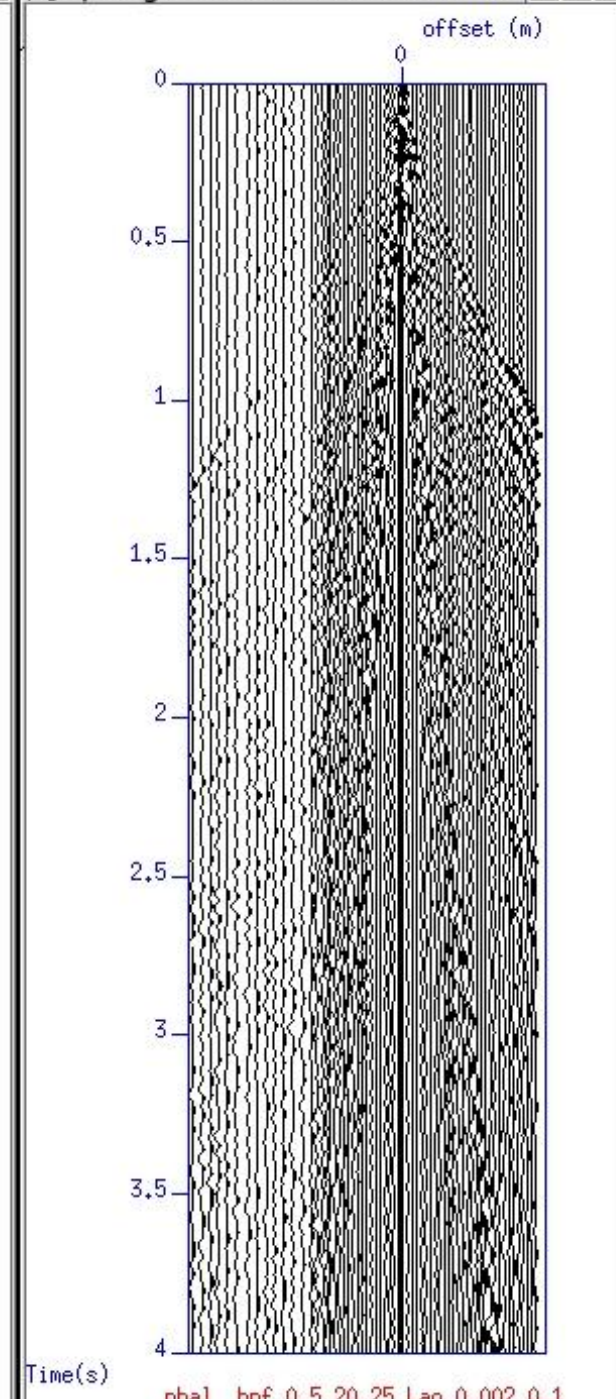
Significant arrivals in SP 4B (191207_2)

- Refractions (mainly)

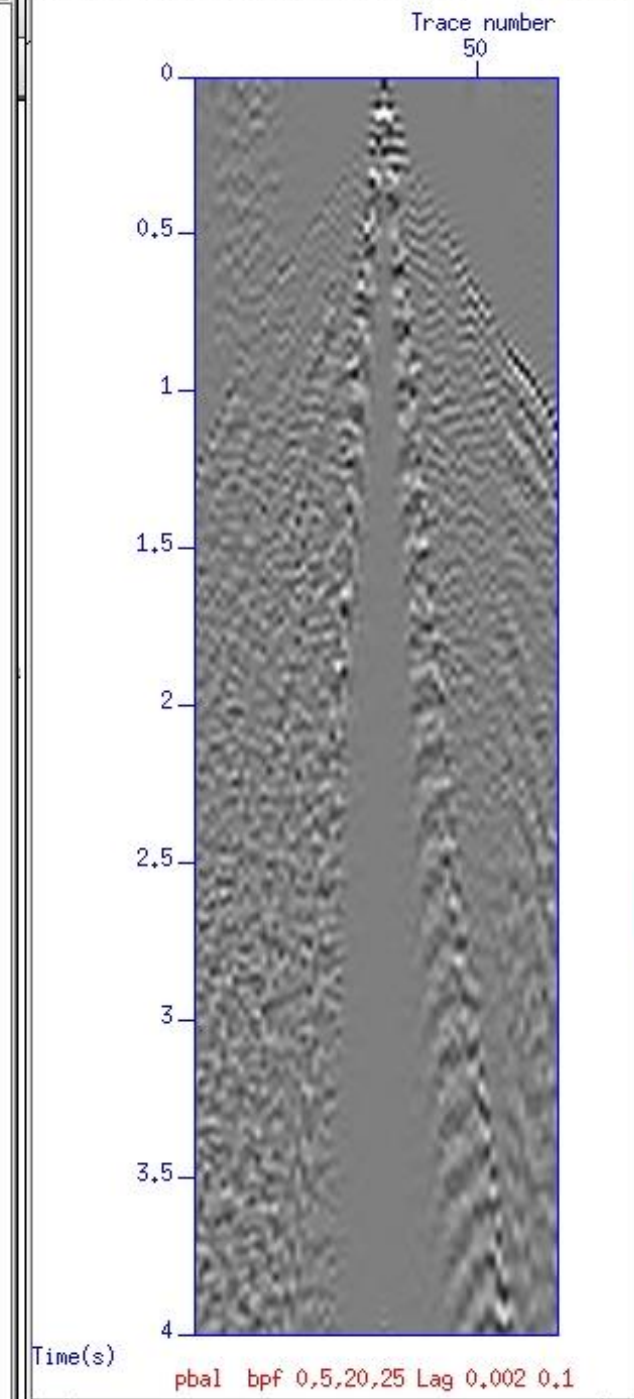
X Trace look -10000_10000 191207_2 <2>



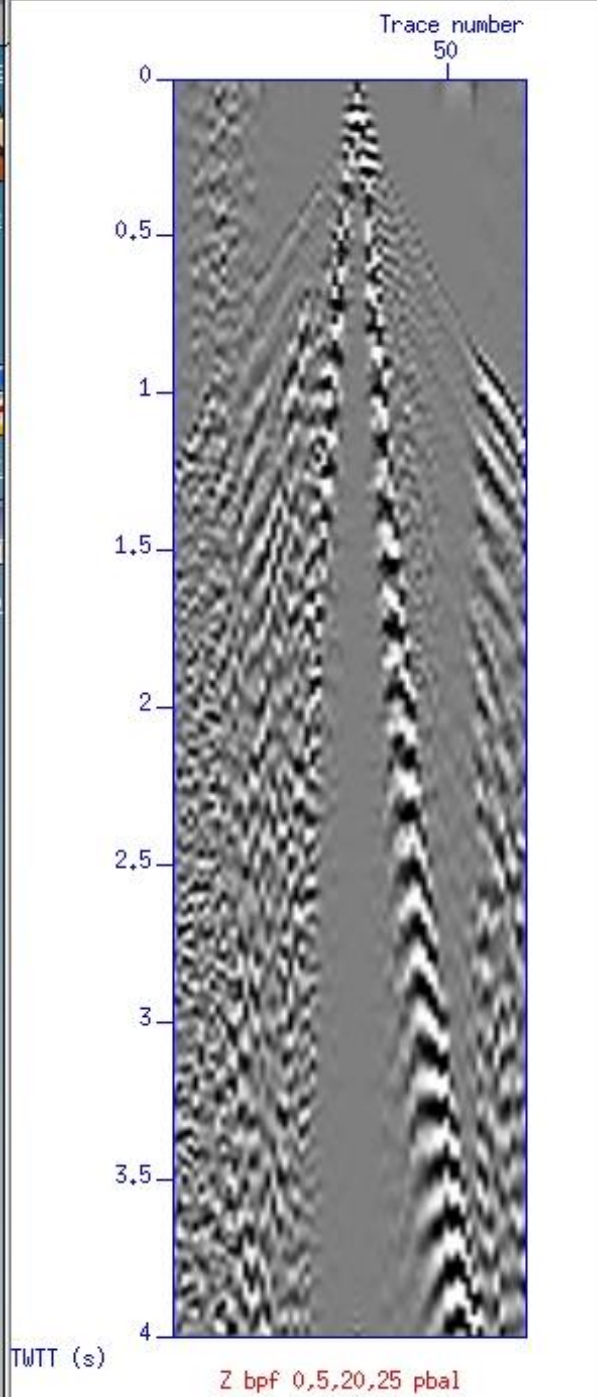
X Spiking Deconvolution191207_2 <2>



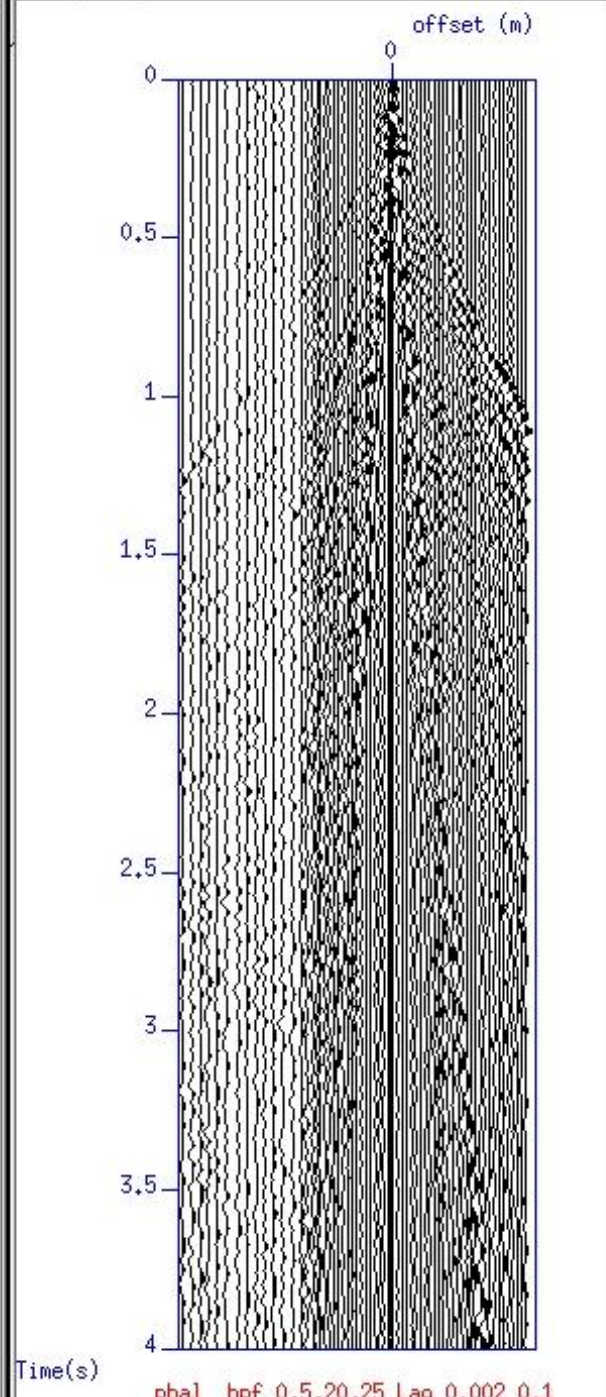
X Spiking Deconvolution191207_2



X Trace look -10000_10000 191207_2 <2>



X Spiking Deconvolution191207_2 <2>



X Spiking Deconvolution191207_2

